Joseph Casey

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EDUCATION

University of Alabama at Birmingham — Birmingham, AL

Ph.D. in Applied Mathematics, **Minor in Computer Science,** (Expected Dec 2026)

- Dissertation: "*Representative Learning for Distributed Support Vector Machines*"
- Research areas: statistics, distributed optimization, support vector machines

M.S. in Mathematics, August 2023

Auburn University — Auburn, AL

Bachelor of Science in Mathematics, 2020

- Minor in Asian Studies with a focus on Chinese

TECHNICAL SKILLS

Programming & Libraries: Python, SQL, NumPy, SciPy, pandas, scikit-learn, PyTorch

Machine Learning: support vector machines, distributed/federated learning, neural networks, natural language processing, large language models

Mathematics: optimization, numerical analysis, probability, statistics, linear algebra

Tools: Git/Github, Jupyter, Anaconda, LaTeX, PostgreSQL, Hugging Face Transformers

RESEARCH / TECHNICAL EXPERIENCE

Doctoral Researcher — Machine Learning & Distributed Optimization, UAB, 2021–2026

- Developed a representative learning framework for distributed SVMs using compact, privacy-retaining summaries from clients.

- Designed and implemented representative learning methods for linear and kernel SVMs, benchmarking against FedAvg, FedProx, and ADMM.
- Derived theoretical guarantees for representative learning methods by analyzing gradient discrepancies, heterogeneity, smoothness, convexity, and convergence.
- Built Python simulation and evaluation pipelines across synthetic and real-world datasets, demonstrating improved performance over distributed learning baselines under severe data heterogeneity and infrequent inter-node communication.
- Conducted additional research in distributed collaborative learning and representative knowledge sharing, resulting in a peer-reviewed publication in *Mathematics*, 2025.

PROFESSIONAL EXPERIENCE

Graduate Teaching Assistant | University of Alabama at Birmingham | *2021–2026*

- Taught six undergraduate courses spanning algebra and precalculus through Calculus I–II.

Marketing Data Analyst Intern | Infinity Insurance (now Kemper) | *2018*

- Analyzed business and customer data and collaborated with programmers and data scientists to identify location-based targets for marketing strategy.
- Used Python, SQL, Excel, Google Analytics, and Google Ads for marketing analysis and reporting across Oracle and MySQL data environments.

TECHNICAL PROJECT

Federated Knowledge Distillation for Biomedical NLP | *PyTorch · BioBERT*

- Built a federated teacher–student knowledge distillation framework using BioBERT representations, KL-divergence/cross-entropy training, and FedAvg benchmarking.

AWARDS & PROFESSIONAL SERVICE

- **Outstanding Graduate Teaching Assistant Award** — 2024
- **Peer Reviewer:** IJCNN — 2026 | *Mathematics* — 2025
- **Volunteer:** UAB Math Talent Search | Alabama Ciphering Contest — 2023–25